ENERGIA: USB-C, carga de bateria, reparto de carga y regulador 3.3 V

J1 USB4085-GF-A — VBUS, CC1, CC2, USB_DM, USB_DP, SBU1, SBU2, SHIELD, GND

U1 TP4056 — VBUS, CHRG_N, STDBY_N, VBAT, GND, PROG

J2 — VBAT, GND, BAT+, BAT−

Bateria LiPo soldada a pads.
Usar una con proteccion integrada

Q1 AO3401A — VBUS, VBAT, VSYS

U2 AP2112K-3.3 — VSYS, PWR_EN, VIN, EN, GND, VOUT, 3V3_REG

D3 1N5819HW — VSYS, VBUS

SW1 ENCENDIDO — PWR_EN, VSYS, GND

JP1 3V3 — 3V3_REG, +3V3

R1 5.1k CC1 · R2 5.1k CC2 · C1 10uF VBUS · R3 3k PROG · C2 10uF VBAT · R6 100k VBUS · C3 1uF VSYS · C4 10uF 3V3_REG · R15 100k PWR_EN · R11 1M VBAT · R12 1M BAT_SENSE · C11 100nF BAT_SENSE · R4 1k VBUS · R5 1k VBUS

D1 CARGANDO (azul) — CHRG_N, LED_CHG_A

D2 CARGA COMPLETA (verde) — STDBY_N, LED_FULL_A

Protección contra estática del USB

U5 USBLC6-2P6 — VBUS, USB_DP, USB_DM, GND

Puntos de prueba y banderas de alimentacion

TP1 VBUS · TP2 VBAT · TP3 VSYS · TP4 +3V3 · TP5 GND · TP6 EN · TP7 BTN_MENU · TP8 UART_TX · TP9 UART_RX

PWR_FLAG VBUS · PWR_FLAG VSYS · PWR_FLAG +3V3 · PWR_FLAG GND

PANTALLA (conector PROVISIONAL)

J3 LCD 2.4in ST7789V — +3V3, GND, LCD_SCK, LCD_MOSI, LCD_CS, LCD_DC, LCD_RST, LCD_LEDA, LCD_LEDK

Q2 2N7002 — LCD_BL, LCD_LEDK, GND

R13 10R (por definir) +3V3 / LCD_LEDA · R14 100k LCD_BL / GND

Pines y resistencia de luz de fondo:
se definen al tener el panel en la mano

microSD (SPI)

J4 TF-01A — GND, GND, SD_MISO, SD_CS, SD_MOSI, SD_SCK, DAT0(DO), DAT1(RSV), DAT2(RSV), CDDAT3(CS), CMD(DI), CLK(SCLK), VDD, CD, VSS, +3V3, GND, GND

R8 10k +3V3 / SD_CS · R9 10k +3V3 / SD_MISO · C8 100nF +3V3 / GND · C12 10uF +3V3 / GND

ESP32-S3-WROOM-1-N16R8

U3 ESP32-S3-WROOM-1-N16R8 — +3V3, GND, EN, BTN_MENU, BAT_SENSE, BTN_L, BTN_X, BTN_UP, BTN_DOWN, BTN_LEFT, BTN_RIGHT, BTN_A, LCD_DC, LCD_CS, LCD_MOSI, LCD_SCK, LCD_BL, LCD_RST, SD_MOSI, SD_SCK, UART_TX, UART_RX, SD_MISO, SD_CS, USB_DM, USB_DP, BTN_B, BTN_R, BTN_OPTION, I2S_BCLK, I2S_LRCLK, I2S_DIN, BTN_Y, BTN_START, BTN_SELECT

SW2 RESET — EN, GND

R7 10k +3V3 / EN · C5 1uF EN / GND · C6 10uF +3V3 / GND · C7 100nF +3V3 / GND

BOTONES (cada uno de su GPIO a GND, pull-up interno)

SW3 UP — BTN_UP, GND · SW4 DOWN — BTN_DOWN, GND · SW5 LEFT — BTN_LEFT, GND · SW6 RIGHT — BTN_RIGHT, GND

SW7 A — BTN_A, GND · SW8 B — BTN_B, GND · SW9 X — BTN_X, GND · SW10 Y — BTN_Y, GND

SW11 START — BTN_START, GND · SW12 SELECT — BTN_SELECT, GND · SW13 MENU — BTN_MENU, GND · SW14 OPTION — BTN_OPTION, GND

SW15 L (gatillo lateral) — BTN_L, GND · SW16 R (gatillo lateral) — BTN_R, GND

MENU (GPIO 0): presionado al encender entra en modo de grabacion.
Botones frontales: solo se usan dos pads en diagonal. L y R son de actuacion lateral.

AUDIO (I2S)

U4 MAX98357A — VSYS, I2S_DIN, I2S_BCLK, I2S_LRCLK, AMP_SD, SD_MODE, GAIN_SLOT, GND, GND, VDD, DIN, BCLK, LRCLK, OUTP, OUTN, EP_PAD, SPK_P, SPK_N

J5 BOCINA+ / BOCINA− — SPK_P, SPK_N

R10 1M VSYS / AMP_SD · C9 10uF VSYS / GND · C10 100nF VSYS / GND

GAIN sin conectar = 9 dB

Borrador generado. Sin probar en hardware.

Sheet: /
File: magsafe-console.kicad_sch

Title: MagSafe Console

Size: A2 | Date: | **Rev: 0.1**
KiCad E.D.A. 9.0.9 | Id: 1/1